

Multi-Stream Convolutional Neural Networks for Fall Detection on the KFall Dataset Using Continuous Wavelet Transform Scalograms

Jaydan Herrera

May 2026

Problem Definition

- Statement and Motivation
 - “Falls are the leading cause of injury for adults ages 65 years and older” (CDC); therefore, this project leverages scalogram representations of inertial measurement unit (IMU) data derived from the KFall Dataset using the Continuous Wavelet Transform (CWT) and applies multi-stream Convolutional Neural Networks (CNNs) to capture temporal–frequency patterns for fall detection.
- Objectives
 - Precision for Fall-class $\geq 85\%$
 - Recall for Fall-class $\geq 80\%$
 - PR AUC for Fall-class $\geq 85\%$
 - Macro F1-score $\geq 85\%$
 - Overall Accuracy $\geq 90\%$

Background

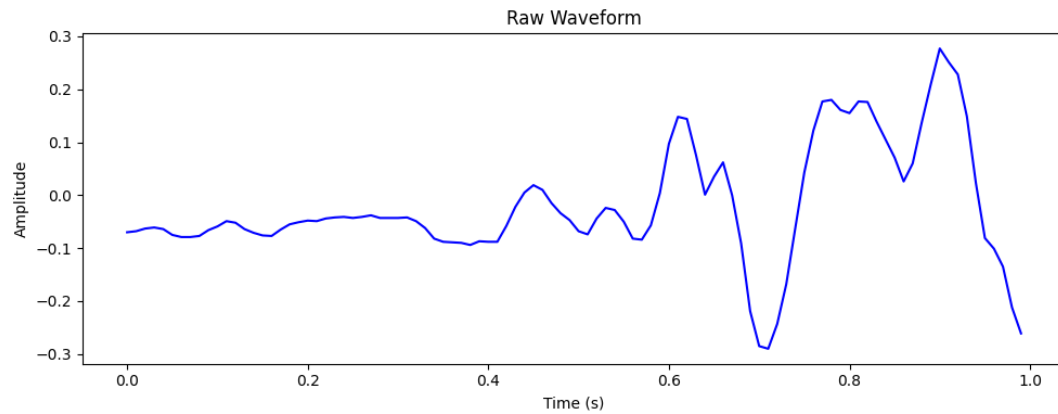
IMU

- Electronic device that uses multiple sensors to measure motion and orientation in three directions (X, Y, Z)
- Typically contains:
 - Accelerometer
 - Measures linear acceleration
 - Gyroscope
 - Measures angular velocity
 - Magnetometer
 - Measures the strength and direction of Earth's magnetic field

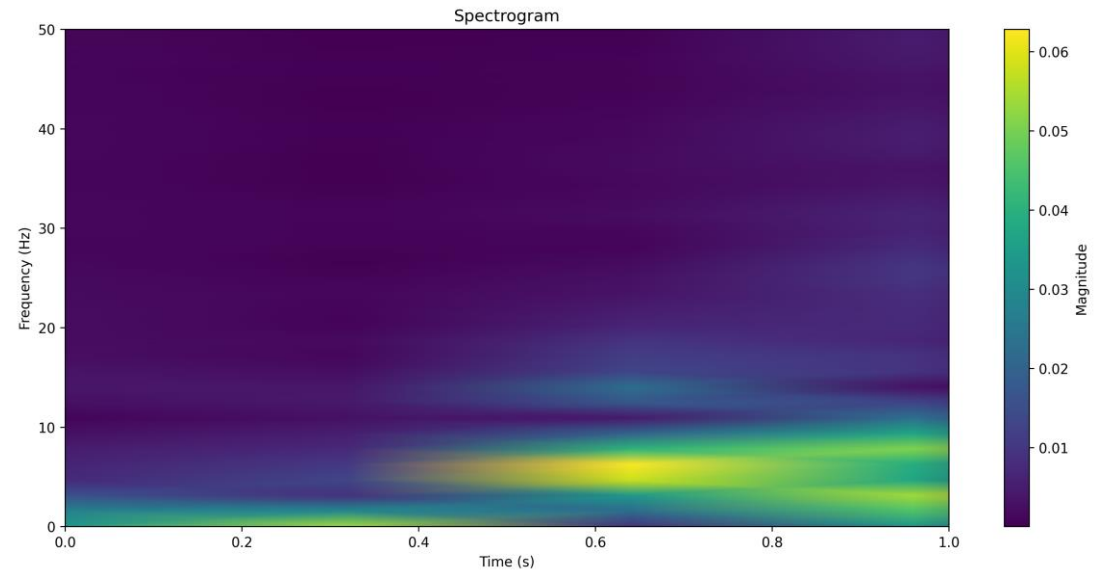


Short-Time Fourier Transform (STFT)

Raw Waveform

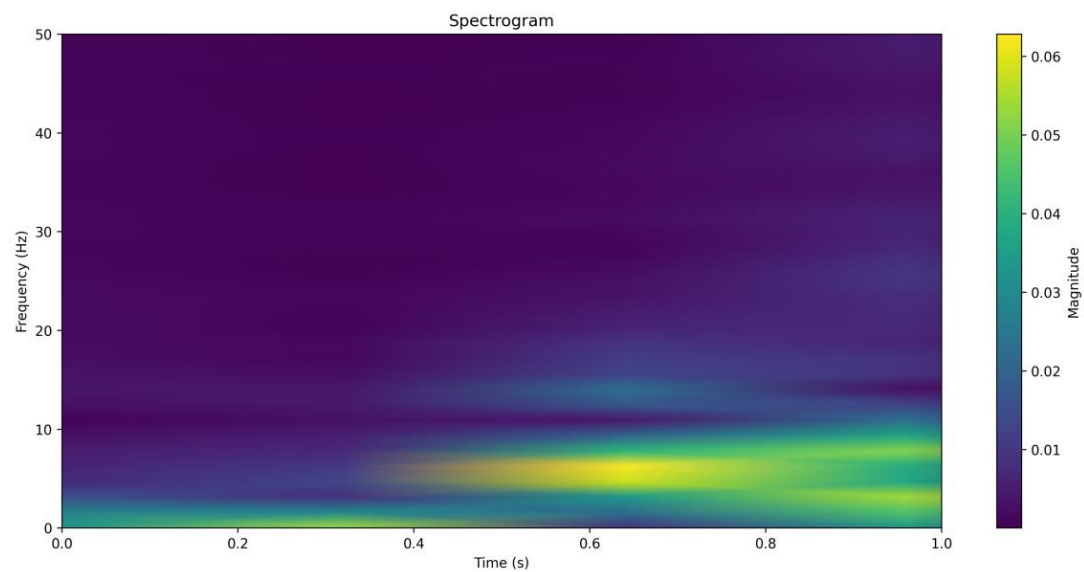


STFT

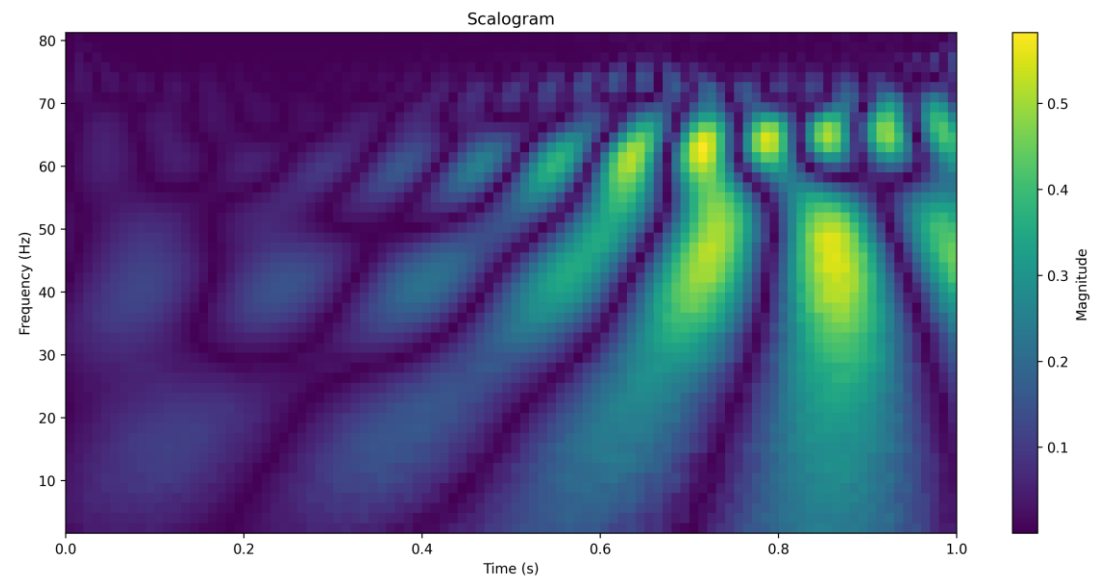


CWT

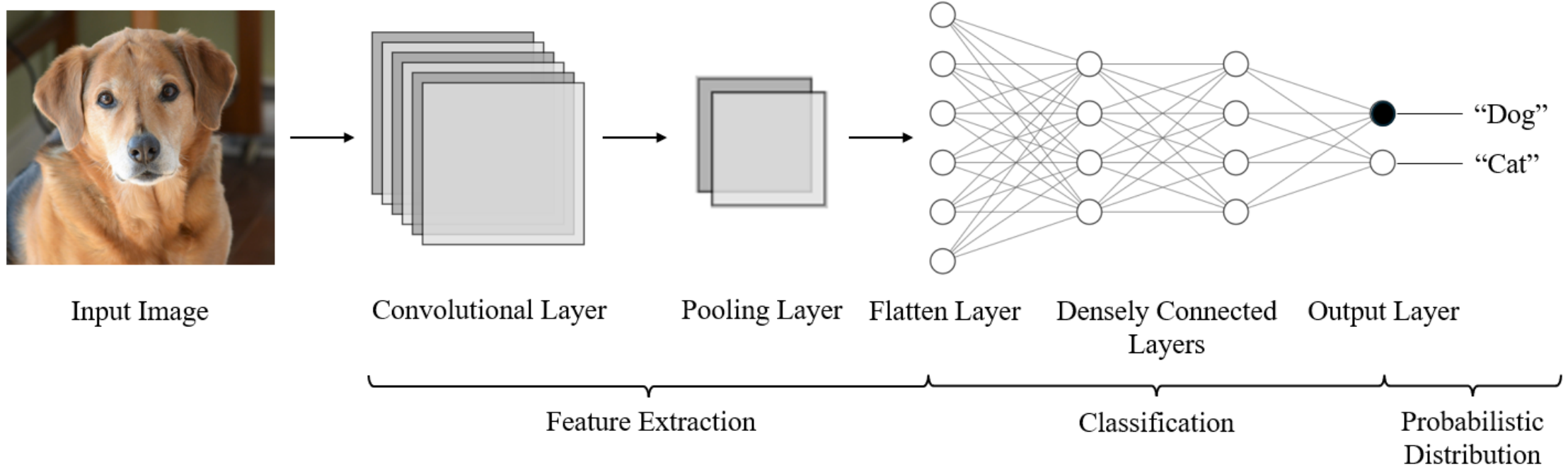
STFT



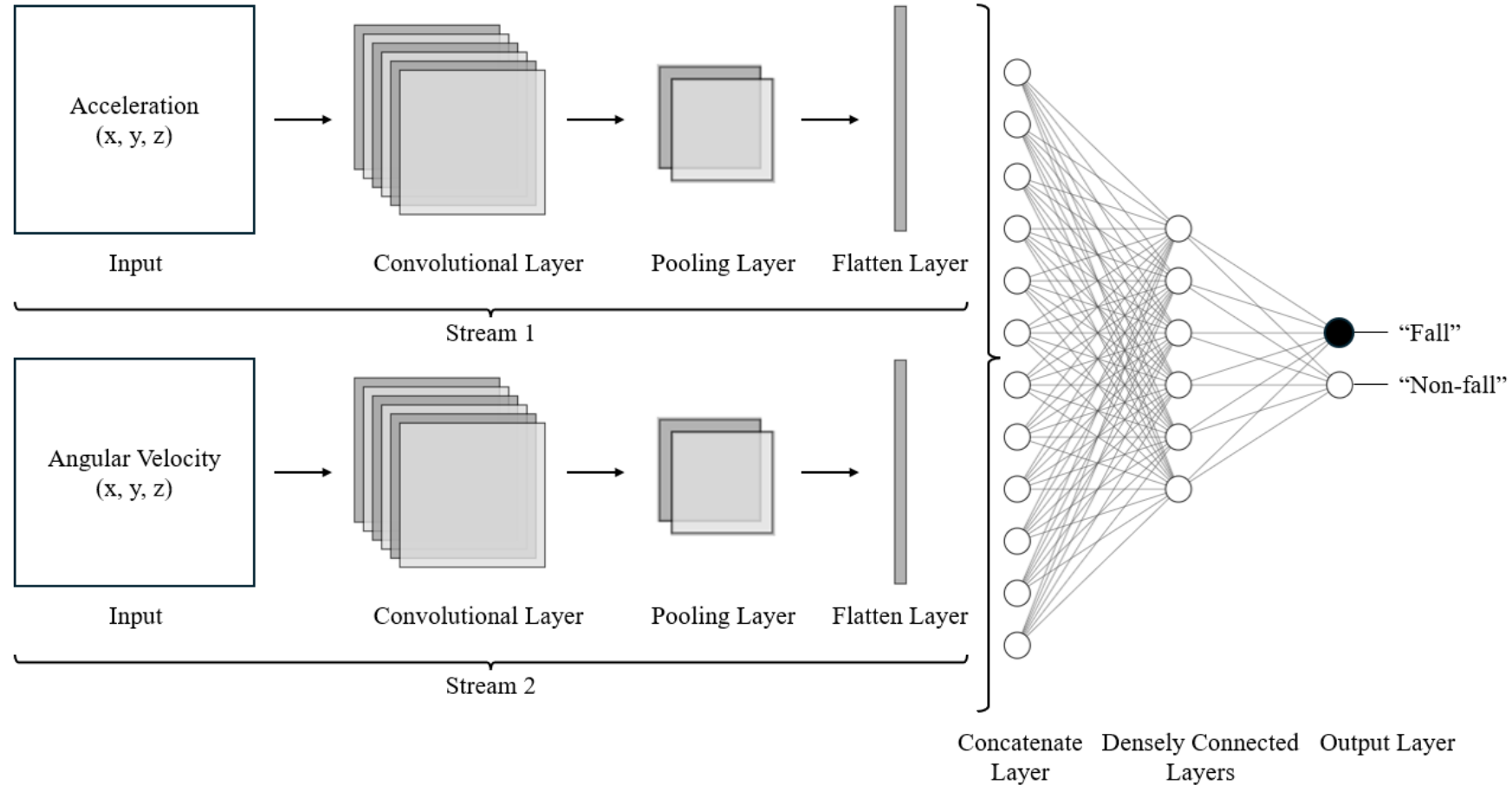
CWT



CNNs



Multi-Stream CNNs

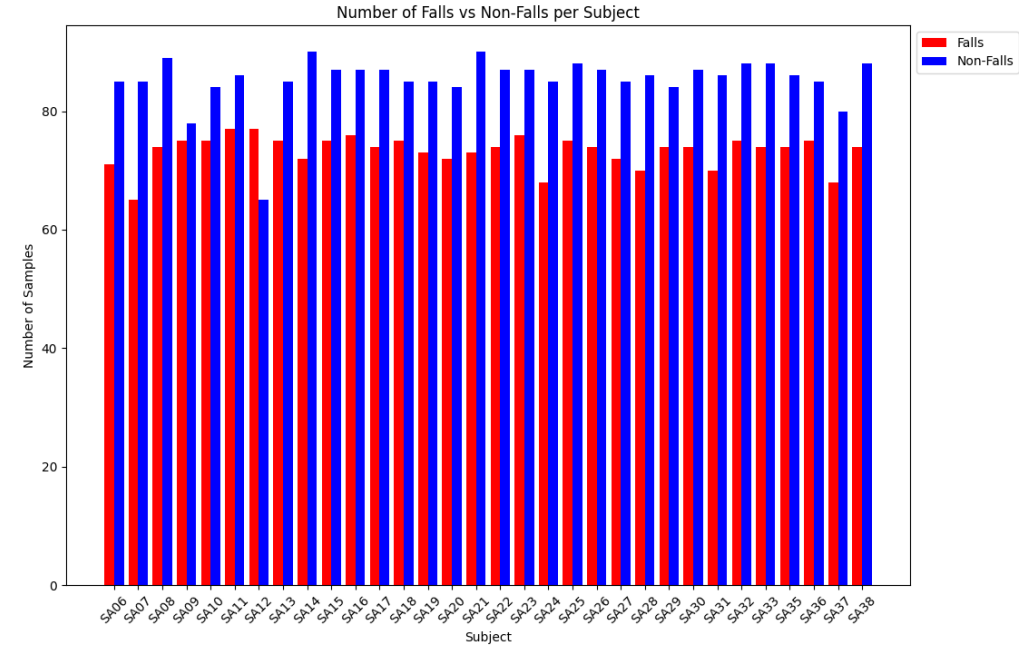
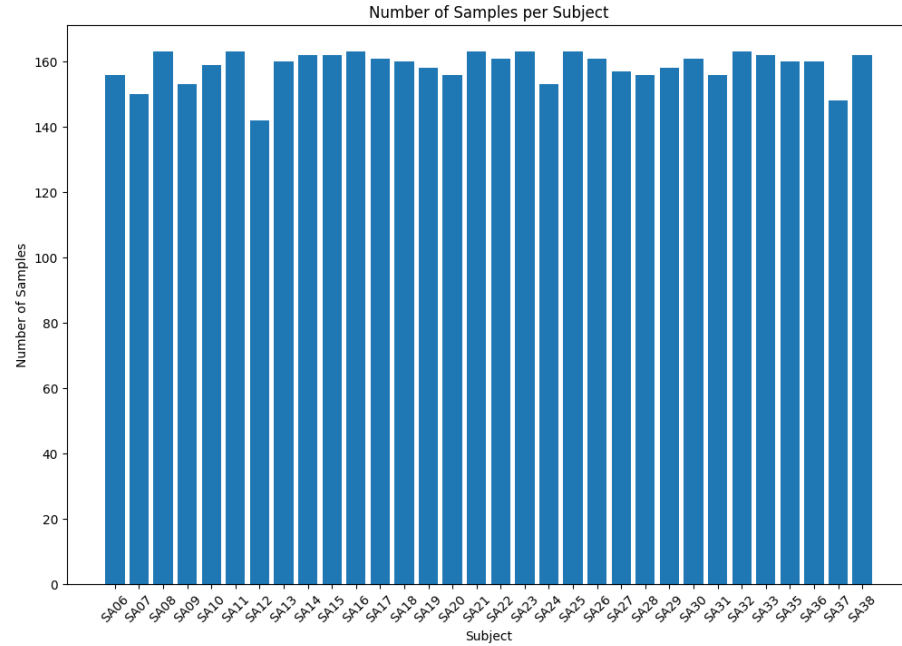


KFall Dataset

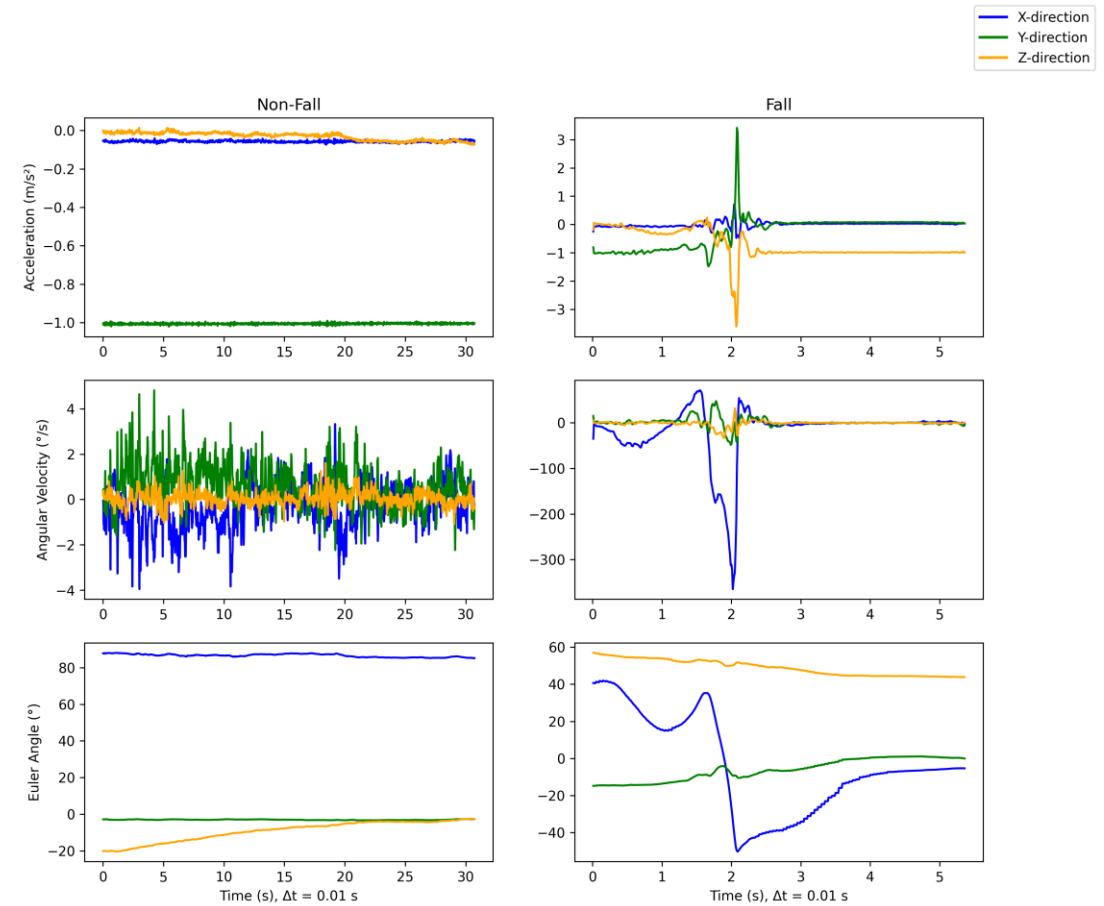
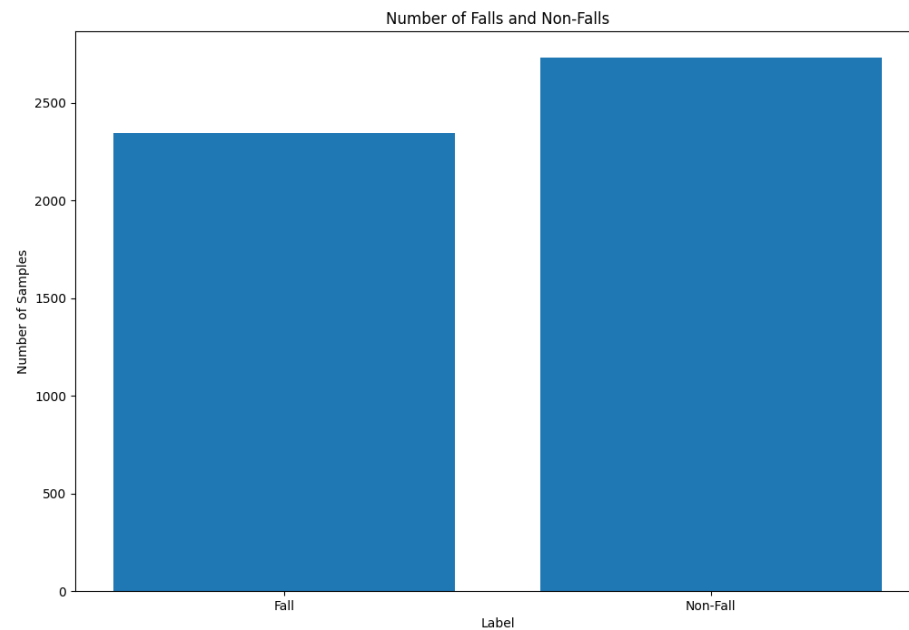
- 32 participant folders
- 5,075 motion files
 - Timestamp
 - Linear acceleration (X, Y, Z)
 - Angular velocity (X, Y, Z)
 - Euler angles (X, Y, Z)
- Fall and non-fall activities
- Metadata provided for labeling

TimeStam	FrameCou	AccX	AccY	AccZ	GyrX	GyrY	GyrZ	EulerX	EulerY	EulerZ
0.01	1	-0.25	-0.809	-0.188	-35.1223	14.9542	4.755551	40.6571	-14.7823	56.95203
0.02	2	-0.07	-1.023	0.056	-2.57831	-0.17189	-0.11459	40.62845	-14.7823	56.9463
0.03	3	-0.07	-1.019	0.056	-3.78152	0.859437	-0.28648	40.58834	-14.7766	56.95203
0.04	4	-0.07	-1.019	0.044	-5.38581	1.948057	-0.45837	40.53105	-14.7594	56.95775
0.05	5	-0.069	-1.025	0.028	-6.53172	2.578311	-0.68755	41.31027	-14.6563	56.53395
0.06	6	-0.067	-1.025	0.021	-7.21927	2.807494	-0.91673	41.23006	-14.6276	56.55095
0.07	7	-0.06	-1.009	0.022	-7.73493	2.922086	-0.74485	41.14984	-14.6047	56.56241
0.08	8	-0.053	-0.995	0.025	-7.39116	2.005353	-0.0573	41.06963	-14.5875	56.5796
0.09	9	-0.049	-0.997	0.025	-7.50575	0.744845	0.572958	40.98942	-14.5875	56.58533
0.1	10	-0.052	-1.002	0.02	-7.90682	-0.34377	1.08862	41.76291	-14.4958	56.17853
0.11	11	-0.06	-1.003	0.017	-8.47978	-0.91673	1.203212	41.67696	-14.5073	56.18426
0.12	12	-0.072	-1.006	0.012	-8.59437	-0.80214	1.145916	41.59102	-14.5245	56.18426
0.13	13	-0.078	-1.015	0.011	-8.76626	-0.45837	1.260508	41.49935	-14.5359	56.18999
0.14	14	-0.081	-1.025	0.009	-9.11003	0	1.260508	41.40195	-14.5474	56.20145
0.15	15	-0.077	-1.032	0	-10.0268	0.343775	1.489691	41.95198	-14.4844	55.89205
0.16	16	-0.072	-1.026	-0.008	-11.3446	0.171887	1.718874	41.83739	-14.4958	55.90924
0.17	17	-0.07	-1.015	-0.009	-12.1467	-0.11459	1.661578	41.70561	-14.513	55.9207
0.18	18	-0.068	-1.006	-0.005	-12.4905	-0.45837	1.546987	41.57956	-14.5302	55.92643
0.19	19	-0.069	-1.006	-0.003	-13.4645	-1.20321	1.375099	41.44205	-14.5474	55.92643
0.2	20	-0.076	-1.01	-0.006	-14.6677	-1.71887	1.375099	41.74572	-14.5245	55.69725
0.21	21	-0.083	-1.01	-0.009	-15.4699	-1.83347	1.604282	41.59102	-14.5531	55.69725
0.22	22	-0.084	-1.005	0	-16.6158	-1.3178	1.604282	41.42486	-14.5761	55.70298
0.23	23	-0.086	-1	0.003	-17.5325	-0.51566	1.317803	41.24152	-14.5875	55.70871
0.24	24	-0.09	-1.003	0.011	-17.1887	0.974029	1.031324	41.0639	-14.5932	55.72017
0.25	25	-0.09	-1.01	-0.001	-17.819	1.948057	0.68755	41.28162	-14.5474	55.53109
0.26	26	-0.083	-1.004	-0.033	-18.9649	1.890761	0.458366	41.08682	-14.5417	55.54828
0.27	27	-0.078	-0.986	-0.043	-18.793	1.317803	-0.17189	40.89201	-14.5359	55.55974
0.28	28	-0.072	-0.969	-0.044	-18.6211	0.744845	-1.20321	40.70294	-14.5245	55.55401
0.29	29	-0.06	-0.954	-0.047	-18.8503	-0.11459	-1.83347	40.51386	-14.513	55.53682
0.3	30	-0.053	-0.952	-0.038	-19.2514	-0.74485	-1.94806	40.66856	-14.4615	55.35347
0.31	31	-0.057	-0.979	-0.018	-21.9443	-2.29183	-0.91673	40.45656	-14.4729	55.33055
0.32	32	-0.058	-1.018	-0.003	-26.4134	-3.15127	0.229183	40.193	-14.5016	55.31337
0.33	33	-0.065	-1.04	-0.007	-29.2209	-2.57831	1.145916	39.89507	-14.5359	55.30191
0.34	34	-0.07	-1.04	-0.028	-31.7992	-2.6929	1.375099	39.57421	-14.5703	55.29618
0.35	35	-0.073	-1.02	-0.046	-34.4921	-2.29183	0.68755	39.44816	-14.5703	55.17586

Data Exploration



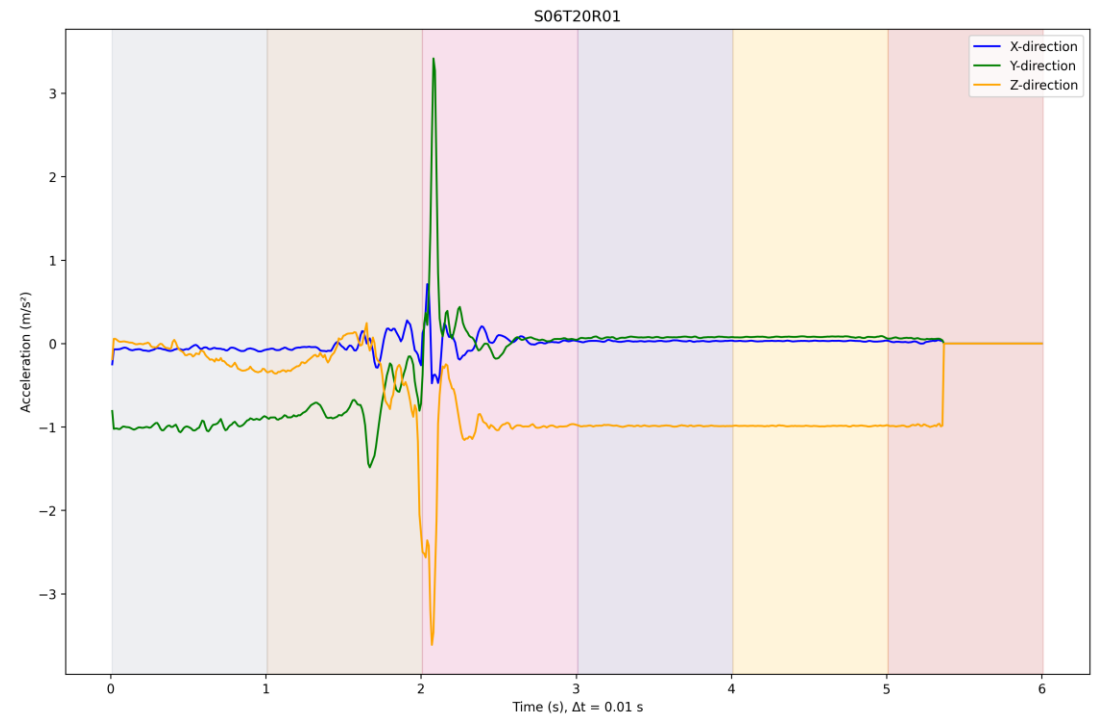
Data Exploration (Cont'd)



Methodology

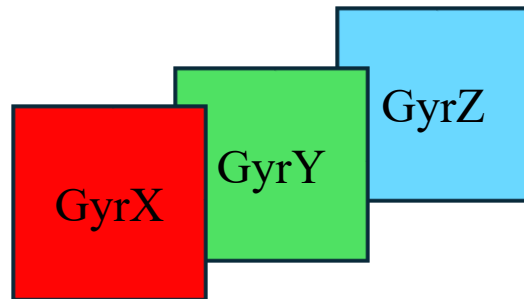
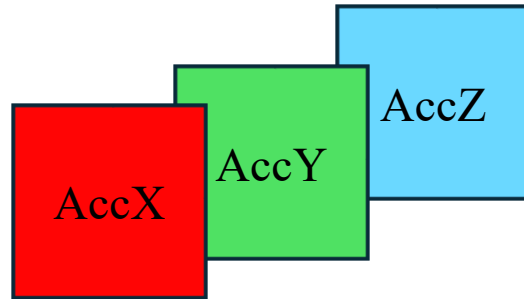
Preprocessing

- Removed Euler angle channels
- Each motion file was segmented into 1-second windows
 - Shorter sequences were padded
- Each segment was labeled as fall or non-fall
- Yielding 42,465 segments
 - 100 samples each
 - 607 fall
 - 5,763 non-fall
 - 6 channels representing
 - Linear acceleration (X, Y, Z)
 - Angular velocity (X, Y, Z)

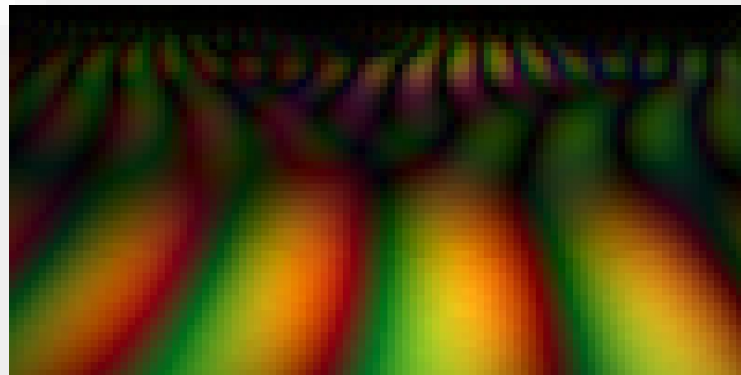
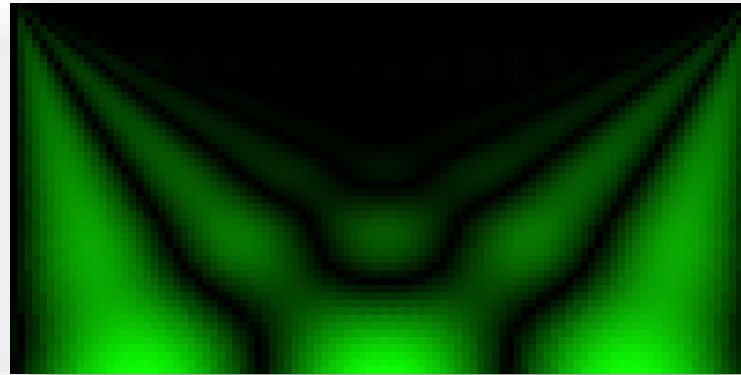


Feature Extraction

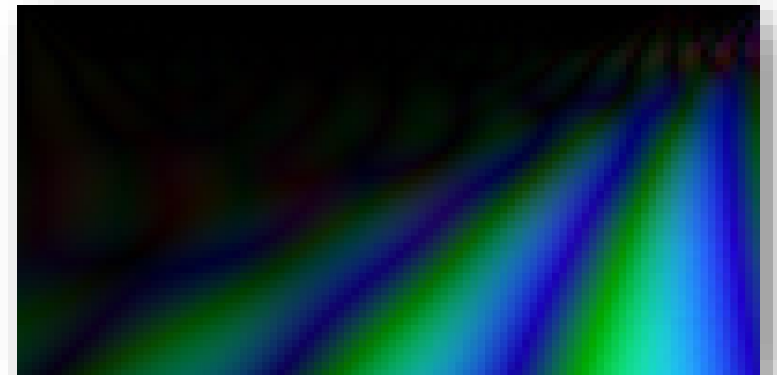
Modality



Non-Fall



Fall



Model Selection

```
# Input shape
input_shape = (50, 100, 3)

# Stream 1
input1 = Input(shape=input_shape)
x1 = layers.Conv2D(32, (3, 3), padding='same')(input1)
x1 = layers.Activation('relu')(x1)
x1 = layers.MaxPooling2D((2, 2))(x1)
x1 = layers.Flatten()(x1)

# Stream 2
input2 = Input(shape=input_shape)
x2 = layers.Conv2D(32, (3, 3), padding='same')(input2)
x2 = layers.Activation('relu')(x2)
x2 = layers.MaxPooling2D((2, 2))(x2)
x2 = layers.Flatten()(x2)

# Merge output from two streams
merged = Concatenate()([x1, x2])

# Densely Connected Layers and Output Layer
x = layers.Dense(128, activation='relu')(merged)
output = layers.Dense(1, activation='sigmoid')(x)

model = Model(inputs=[input1, input2], outputs=output)
```

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 50, 100, 3)]	0	[]
input_2 (InputLayer)	[(None, 50, 100, 3)]	0	[]
conv2d (Conv2D)	(None, 50, 100, 32)	896	['input_1[0][0]']
conv2d_1 (Conv2D)	(None, 50, 100, 32)	896	['input_2[0][0]']
activation (Activation)	(None, 50, 100, 32)	0	['conv2d[0][0]']
activation_1 (Activation)	(None, 50, 100, 32)	0	['conv2d_1[0][0]']
max_pooling2d (MaxPooling2D)	(None, 25, 50, 32)	0	['activation[0][0]']
max_pooling2d_1 (MaxPooling2D)	(None, 25, 50, 32)	0	['activation_1[0][0]']
flatten (Flatten)	(None, 40000)	0	['max_pooling2d[0][0]']
flatten_1 (Flatten)	(None, 40000)	0	['max_pooling2d_1[0][0]']
concatenate (Concatenate)	(None, 80000)	0	['flatten[0][0]', 'flatten_1[0][0]']
dense (Dense)	(None, 128)	10240128	['concatenate[0][0]']
dense_1 (Dense)	(None, 1)	129	['dense[0][0]']

=====

Total params: 10,242,049
Trainable params: 10,242,049
Non-trainable params: 0

Model Training

- Dataset split
 - 70% training, 15% validation, 15% testing using stratified sampling
- Optimizer
 - Adam with initial learning rate of 0.001
- Learning rate scheduling
 - Reduced by 50%, if validation loss does not improve for 3 epochs, and minimum learning rate of 0.000001
- Learning rate tracking
 - Recorded at the end of each epoch using custom callback
- Loss function
 - Binary cross-entropy
- Training setup
 - 50 epochs with batch size of 32
- Monitoring metric
 - Accuracy

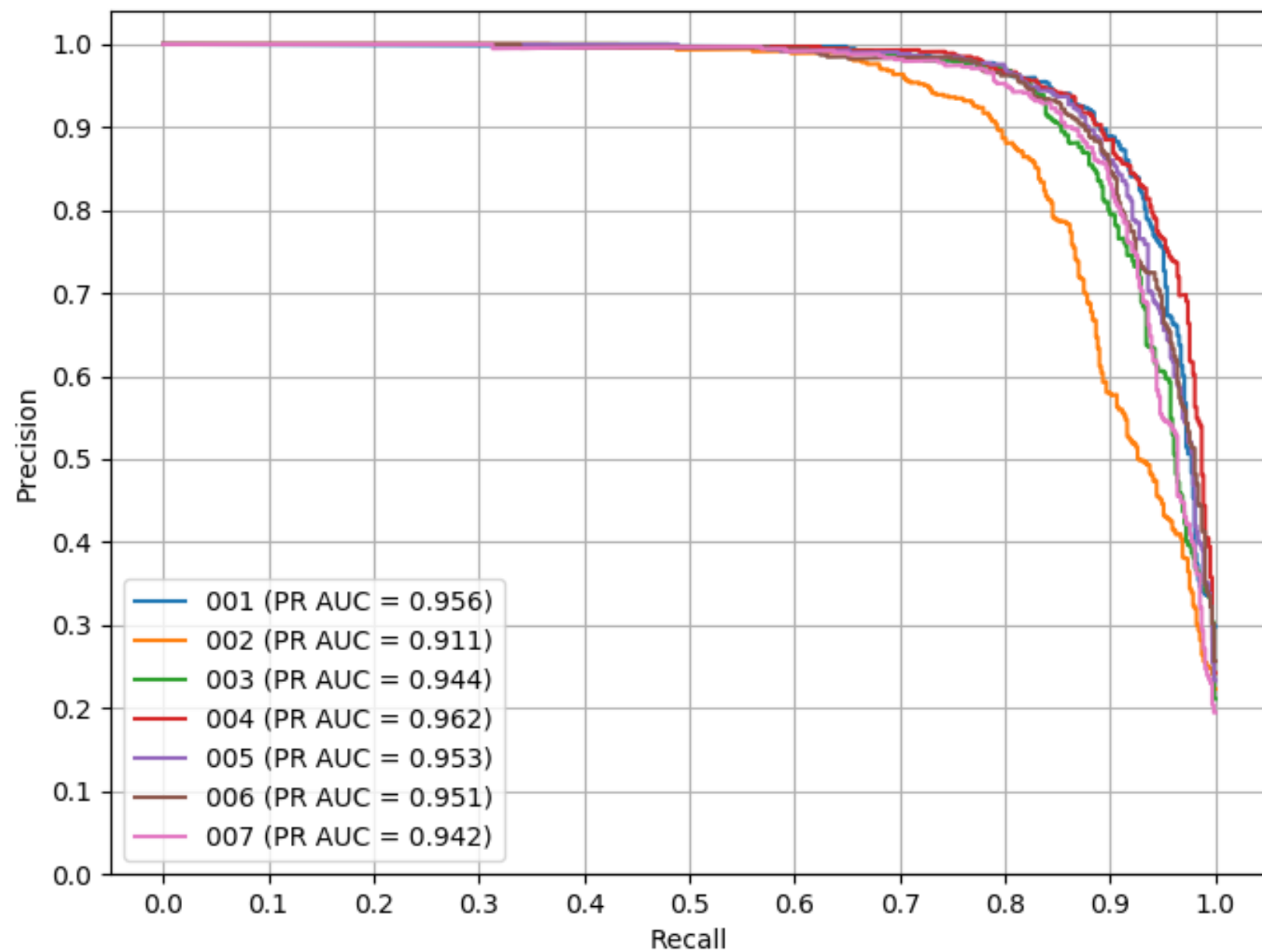
Model Evaluation

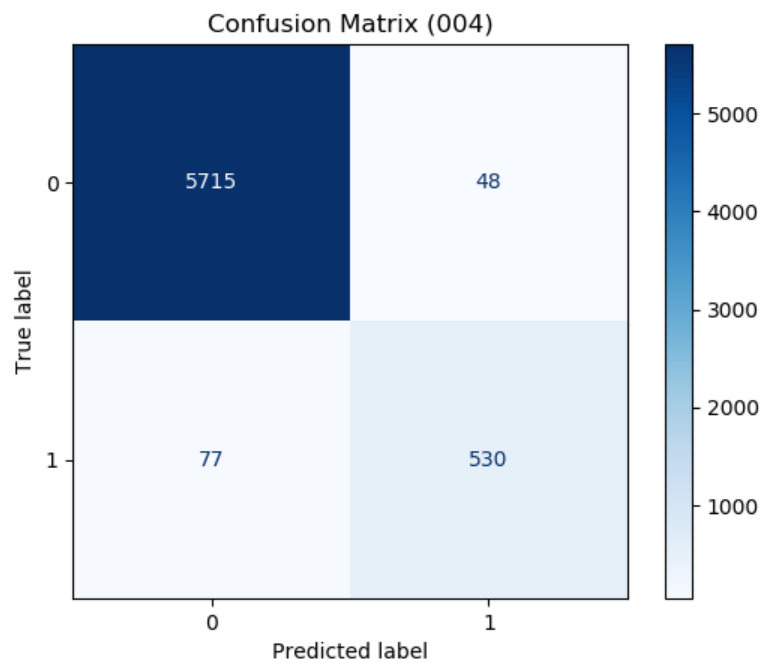
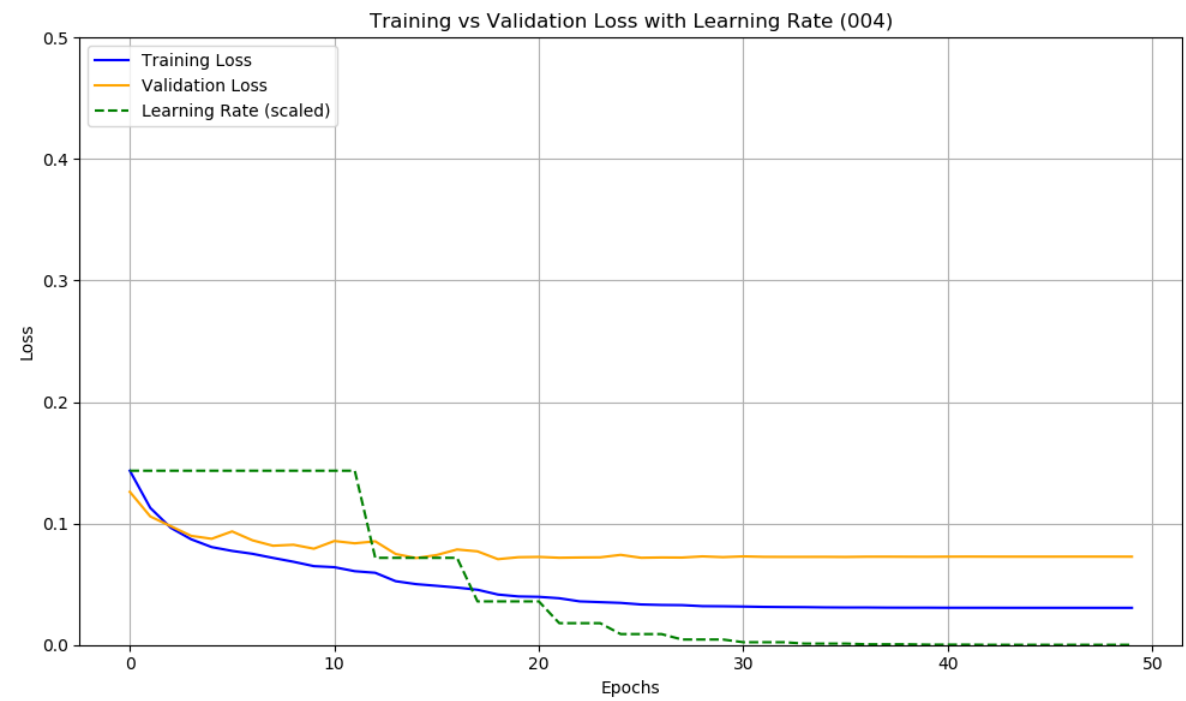
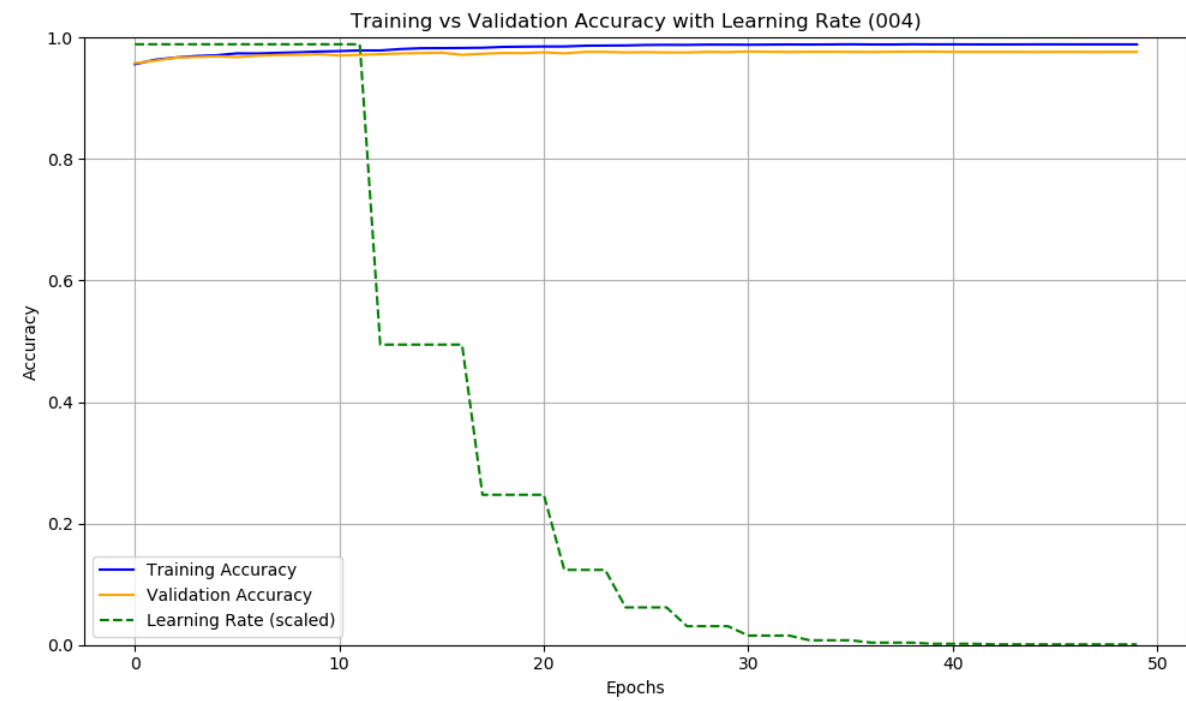
- Evaluated models on test set (15% of dataset)
- Generated:
 - Classification reports:
 - Accuracy
 - Precision
 - Recall
 - F-1 score
 - Macro Average
 - Weighted Average
 - Confusion matrices
 - Precision-recall curves

Results

Experiment ID	Configuration	Trainable Parameters	Non-Trainable Parameters	Total Parameters	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
001	Base Model	10,242,049	0	10,242,049	0.9984	0.0053	0.9765	0.1247
002	Replaced max pooling and flatten layers with global average pooling layers.	10,241	0	10,241	0.9733	0.0766	0.9699	0.0924
003	Added one convolutional block per stream, 32 filters each	28,737	0	28,737	0.9795	0.0579	0.9749	0.0797
004	Added convolutional block to each stream	47,233	0	47,233	0.9889	0.0306	0.9766	0.0727
005	Removed one convolutional block from each stream and updated filters to (32, 64)	55,425	0	55,425	0.9856	0.0407	0.9755	0.0778
006	Reduced filters to (32, 32) and added dropout to fully connected layers	28,737	0	28,737	0.9806	0.0549	0.9755	0.0784
007	Added batch normalization	28,993	256	29,249	0.9756	0.0708	0.9728	0.0841

Precision-Recall Curve for All Models





Classification Report (004)				
-	Precision	Recall	F1-Score	Support
0	0.99	0.99	0.99	5,763
1	0.92	0.87	0.89	607
Accuracy	-	-	0.98	6,370
Macro Average	0.95	0.93	0.94	6,370
Weighted Average	0.98	0.98	0.98	6,370

Conclusion and Future Work

- Conclusion

- CWT scalograms combined with a multi-stream CNN effectively distinguish between falls and non-falls on KFall dataset

Metric	Objective	Result	Percentage Change
Precision (F)	$\geq 85\%$	92%	+8%
Recall (F)	$\geq 80\%$	87%	+8%
PR AUC (F)	$\geq 85\%$	96%	+12%
Macro F1-Score	$\geq 85\%$	95%	+11%
Accuracy	$\geq 90\%$	98%	+8%

- Future work

- Collection of additional data
- Address class imbalance
- Further experimentation of model architectures and hyperparameters
- Custom loss functions
- Comparison of single-modality approaches
- Pruning and quantization for comparison against full model

Questions?



References

- GitHub
 - <https://github.com/jhJokercoder/FallDetection-SeniorProject>
- Sources
 - “LPMS-B2 Series: 9-Axis Inertial Measurement Unit / AHRS with Bluetooth Classic and BLE Connectivity.” *Life Performance Research*, www.lp-research.com/products/inertial-measurement-units-imu/lpms-b2/.
 - Kerns, Nancy. “The Dogs We Love: Too Little Time Together.” *Whole Dog Journal*, 10 July 2018, www.whole-dog-journal.com/blog/the-dogs-we-love-too-little-time-together/.
 - Yu, Xiaoqun, et al. “A Large-Scale Open Motion Dataset (KFall) and Benchmark Algorithms for Detecting Pre-impact Fall of the Elderly Using Wearable Inertial Sensors.” *Frontiers in Aging Neuroscience*, vol. 13, 2021, pp. 1-14. doi.org/10.3389/fnagi.2021.692865.